

The complexity of finding bounded subtrees

Damir D. Dzhafarov
University of Connecticut
www.damirdzhafarov.com

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Bounded subtree principles

Basic question. Given an ill-founded tree $T \subseteq \omega^{<\omega}$, how difficult is it to extract an ill-founded finitely branching subtree $S \subseteq T$?

For example, if we are given a path $f \in [T]$ then we can take

$$T^{\leq f} = \{\sigma \in T : (\forall i < |\sigma|)[\sigma(i) \leq f(i)]\},$$

i.e., the part of T bounded by f .

The question comes (somewhat obliquely) out of Mark Carney's thesis at Leeds, who was looking at certain problems involving Wang tiles under Weihrauch reducibility.

We will be interested in the question for its own sake, using computability and reverse math.

Paths and subtrees

Suppose $T \subseteq \omega^{<\omega}$ is a computable ill-founded tree.

Basic question, restated. What is the relationship between the degrees of the finitely branching subtrees $S \subseteq T$ and the degrees of the paths of T ?

- ▶ If $f \in [T]$ then $T \leq^f$ is computable from f .
- ▶ If $S \subseteq T$ is finitely branching and ill-founded there exists $f \in [T]$ with $f \leq_T S''$.

So there are some similarities:

- ▶ there is always a subtree S computable from Kleene's $\mathcal{O}$;
- ▶ there is a T with no Δ_1^1 subtree S ; etc.

What about coding? (There exist T all of whose paths compute $0^{(\omega)}, \dots$)

What if we ask that S contain a specific path? (E.g., the leftmost path of T ?)

Formalizing subtree principles

Definition. The following definitions are made in RCA_0 :

- ▶ H : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$.
- ▶ H^+ : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$ such that for all $f \in [T] \setminus [S]$ there exists $g \in [T]$ with $g <_{\text{lex}} f$.
- ▶ H^{++} : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$ such that for all $f \in [T] \setminus [S]$ there exists $g \in [S]$ with $g <_{\text{lex}} f$.

Classically, each of H^+ and H^{++} are just asserting that the subtree S contains the lexicographically least (leftmost) path of T . We will see that the slight difference between H^+ and H^{++} makes a big difference.

Clearly, $H^{++} \rightarrow H^+ \rightarrow H$.

The strength of H and H^+

H : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$.

H^+ : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$ such that for all $f \in [T] \setminus [S]$ there exists $g \in [T]$ with $g <_{\text{lex}} f$.

Proposition. H^+ (and hence also H) is provable in RCA_0 .

Proof. Fix an ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$.

If T has a leftmost path f , then setting $S = T^{\leq f}$ satisfies the conclusion of H^+ .
(Recall: $T^{\leq f} = \{\sigma \in T : (\forall i < |\sigma|)[\sigma(i) \leq f(i)]\}$. Thus $f \in [T^{\leq f}]$.)

If T does not have a leftmost path, then every ill-founded finitely branching subtree of T trivially satisfies the conclusion of H^+ .

So in this case, we can let $S = T^{\leq f}$ for any $f \in [T]$.

The strength of H^{++}

H^+ : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$ such that for all $f \in [T] \setminus [S]$ there exists $g \in [T]$ with $g <_{\text{lex}} f$.

H^{++} : Every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has an infinite, finitely branching subtree $S \subseteq T$ such that for all $f \in [T] \setminus [S]$ there exists $g \in [S]$ with $g <_{\text{lex}} f$.

Proposition. Over ACA_0 , H^{++} implies $\Pi_1^1\text{-CA}_0$.

Proof. We show that every ill-founded tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$ has a leftmost path.

Fix such a T , and apply H^{++} to get a finite branching subtree $S \subseteq T$.

Since S is finite branching, ACA_0 proves that S has a leftmost path f .

If f were not the leftmost path in T , there would be some $h \in [T]$ with $h <_{\text{lex}} f$. Then $h \notin [S]$, so there is some $g \in [S]$ with $g <_{\text{lex}} h <_{\text{lex}} f$, contradiction.

Corollary. H^{++} is not provable even in ACA_0 .

Cone avoidance

This naturally leads us to ask whether H^{++} implies ACA_0 .

Theorem. Fix $C \not\leq_T A$. For every A -computable ill-founded tree $T \subseteq \omega^{<\omega}$ and every $f \in [T]$, there exists a finitely branching subtree $S \subseteq T$ such that $f \in [S]$ and $C \not\leq_T A \oplus S$.

Corollary. H^{++} admits cone avoidance. Hence, there is an ω -model of $RCA_0 + H^{++}$ in which ACA_0 fails.

But this also answers our original question about finitely branching subtrees and coding. Thus in general, there is a stark difference between the degrees of the paths of a computable tree $T \subseteq \omega^{<\omega}$, and the degrees of its finitely branching ill-founded subtrees S (even if we ask S to contain a specific path).

Proof of the theorem

Theorem. Fix $C \not\leq_T A$. For every A -computable ill-founded tree $T \subseteq \omega^{<\omega}$ and every $f \in [T]$, there exists a finitely branching subtree $S \subseteq T$ such that $f \in [S]$ and $C \not\leq_T A \oplus S$.

Proof. Take $A = \emptyset$ for simplicity. Fix C , T , and $f \in [T]$.

We force with the following notion of forcing:

- ▶ a **pre-condition** is a pair (D, u) where $D \subseteq T$ is a finite tree, $u \in \omega$, and there is at least one $\sigma \in D$ with $|\sigma| = u$;
- ▶ a **condition** is a pre-condition (D, u) containing $f \upharpoonright u$; (E.g., $(\{\lambda\}, 0)$)
- ▶ a pre-condition (E, v) **extends** (D, u) if $D \subseteq E$ and every $\tau \in E \setminus D$ extends some $\sigma \in D$ of length u .

Conditions don't behave nicely with computations. Turns out to be useful.

Proof of the theorem

A condition is a pair (D, u) where $D \subseteq T$ is a finite tree, $u \in \omega$, and $f \restriction u \in D$.

Let $\mathcal{F}$ be a generic filter. Then $\mathcal{F}$ determines a finitely branching subtree $S \subseteq T$ with $f \in [S]$.
 $(S = \bigcup_{(D,u) \in \mathcal{F}} D.)$

First, suppose that for each $e \in \omega$ the set of conditions (D, u) satisfying either Case 1 or Case 2 below is dense.

- ▶ Case 1. For every n , and all pre-conditions $(E, v), (F, w) \leq (D, u)$ we have that $\Phi_e^F(n) \simeq \Phi_e^E(n)$.
- ▶ Case 2. There is an n such that $\Phi_e^D(n) \downarrow \neq C(n)$ and $\Phi_e^E(n) \downarrow = \Phi_e^D(n)$ for all conditions $(E, v) \leq (D, u)$.

If $\mathcal{F}$ contains a condition satisfying Case 1 then it cannot be that $\Phi_e^S = C$ else $C \leq_T \emptyset$. If $\mathcal{F}$ contains a condition satisfying Case 2 then $\Phi_e^S(n) \downarrow \neq C(n)$.

Proof of the theorem

Now fix $e \in \omega$ and a condition (D, u) with no extension satisfying Case 1 or 2.

We define an computable function g inductively so that $f(x) \leq g(x)$ for all x .
From here, $S = T^{\leq g}$ is a computable finitely branching subtree with $f \in [S]$.

Fix x and suppose the result is true for $f \upharpoonright x$ and $g \upharpoonright x$.

Define $D_x = D \cup \{\sigma \in T : f \upharpoonright u \preceq \sigma \wedge |\sigma| \leq x \wedge (\forall y < x)[\sigma(y) \leq g(y)]\}$.
(This is a condition extending (D, u) .)

- ▶ Since Case 1 fails, there is an n and pre-conditions $(E, v), (F, w) \leq (D_x, x)$ such that $\Phi_e^E(n) \downarrow \neq \Phi_e^F(n) \downarrow$. (Can be found computably.)
- ▶ Let $g(x)$ be the maximum of the uses of the computations $\Phi_e^F(n)$ and $\Phi_e^E(n)$.
- ▶ If we had $f(x) > g(x)$ then $(E \cup \{f \upharpoonright v + 1\}, v + 1)$ or $(F \cup \{f \upharpoonright w + 1\}, w + 1)$ would be an extension of (D, u) witnessing Case 2.

Choice and semi-choice principles

Definition. We define the following instance-solution problems.

- ▶ $C_{\mathbb{N}^{\mathbb{N}}}$: Instances are ill-founded trees $T \subseteq \omega^{<\omega}$;
solutions to T are all $f \in [T]$.
- ▶ F : Instances are ill-founded trees $T \subseteq \omega^{<\omega}$;
solutions to T are all pairs $\langle f, g \rangle$ such that $f \in [T]$ or $g \in [T]$.

Open question. Does every ill-founded tree T uniformly compute an ill-founded tree U such that every pair $\langle f, g \rangle$ with $f \in [U]$ or $g \in [U]$ uniformly computes some $h \in [T]$?
(i.e., is it the case that $C_{\mathbb{N}^{\mathbb{N}}} \leq_w F$?)

- ▶ It is not hard to see that the answer is **no** if we insist that $T, U \subseteq 2^{<\omega}$.
- ▶ For silly reasons, the answer is **yes** if we only insist that $T \subseteq 2^{<\omega}$ (but not U).
(Ensure all paths of U dominate settling time function of T .)

A partial result

Open question. Is it the case that $C_{\mathbb{N}^{\mathbb{N}}} \leq_W F$?

Proposition. For every pair of functionals Φ and Ψ , there exists a computable ill-founded tree $T \subseteq \omega^{<\omega}$ such that if $\Phi(T) = U \subseteq \omega^{<\omega}$ is an ill-founded tree then there exists $f \in [U]$ and a partial function g such that either $\Psi(f, g)$ or $\Psi(g, f)$ is not an element of $[T]$.

The proof draws some elements from the proof of the theorem.

- ▶ Fix a T with no hyperarithmetical paths.
- ▶ Argue that there must be some $\alpha \in U$ so that $\Psi(\alpha, 0^n) = \beta \notin T^{\text{ext}}$.
- ▶ Now α cannot be extendible in U . But also, there can be no x larger than the use of the computation $\Psi(\alpha, 0^n)$ so that $0^n x$ is extendible.
- ▶ Hence, U is compact and thus has a hyperarithmetical path.

Thank you for your attention.
